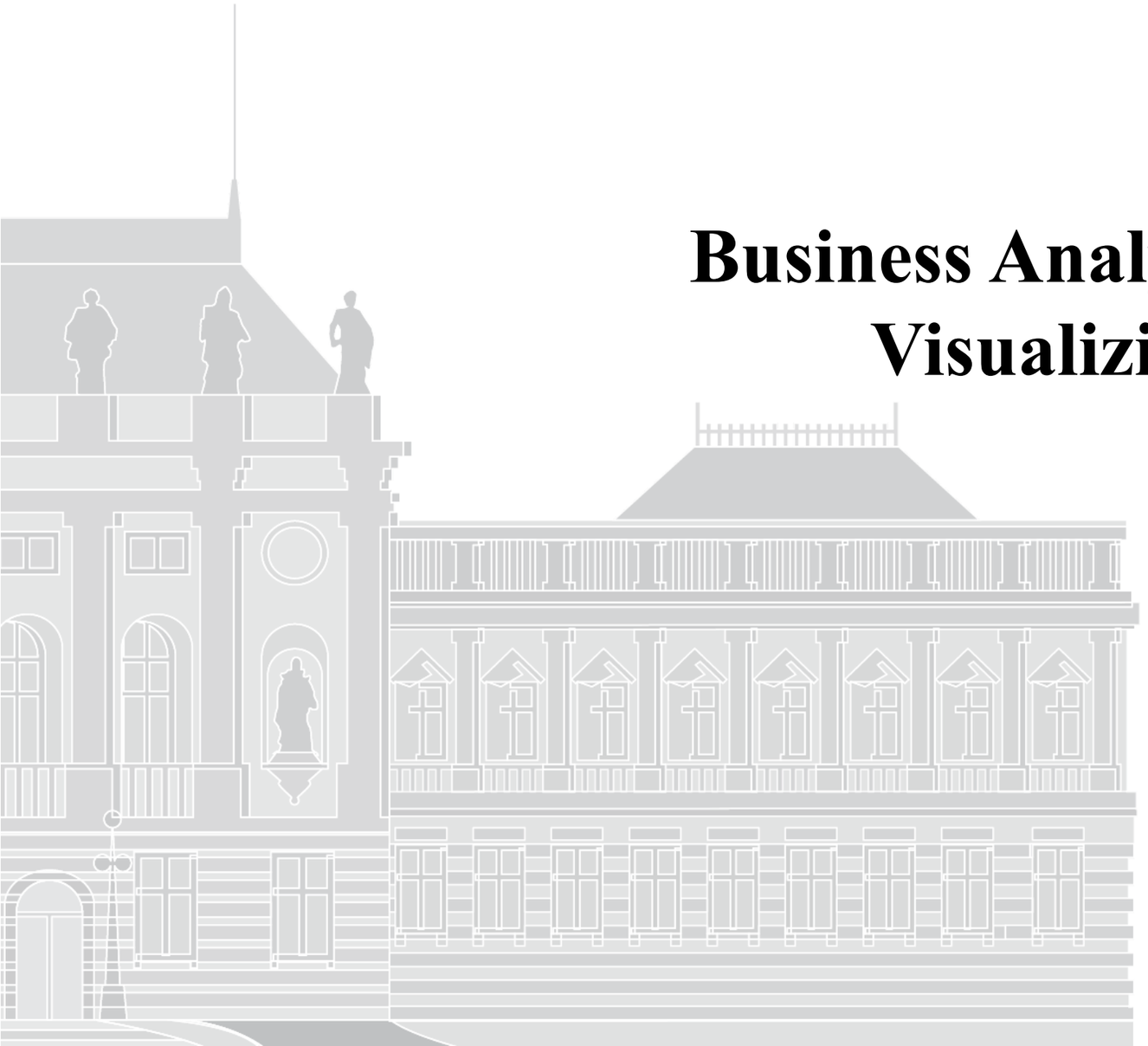
Business Analytics with R – Visualizing Data



Business Analytics with R – Visualizing Data

Outline

01

Why visualize data?

02

Goals of the current lecture

03

What do we want you to do with Data Visualization?

Which insights do we want you to generate?

04

Visualizing Data in R

Goals of the current lecture

Overview through Exploration

Students can gain an overview over data through the use of explorative and graphical techniques.

Context-Dependent Approaches

Data Visualization can have many different uses. Each use case has different requirements and needs to be approached differently. This means you may need different tools, depending on what the exact goal of the project you are working on is.

Focus of This Lecture

The focus of this lecture is on exploring data visually with the goal of using the resulting insights for the development of machine learning models.

Note

You will be able to use the tools, technologies and methods in several other contexts as well.

Why visualize data?

➤ One (or more) of several reasons:

1

Communicating insights to decision makers (i.e. Management, ...)

2

Data Analysis

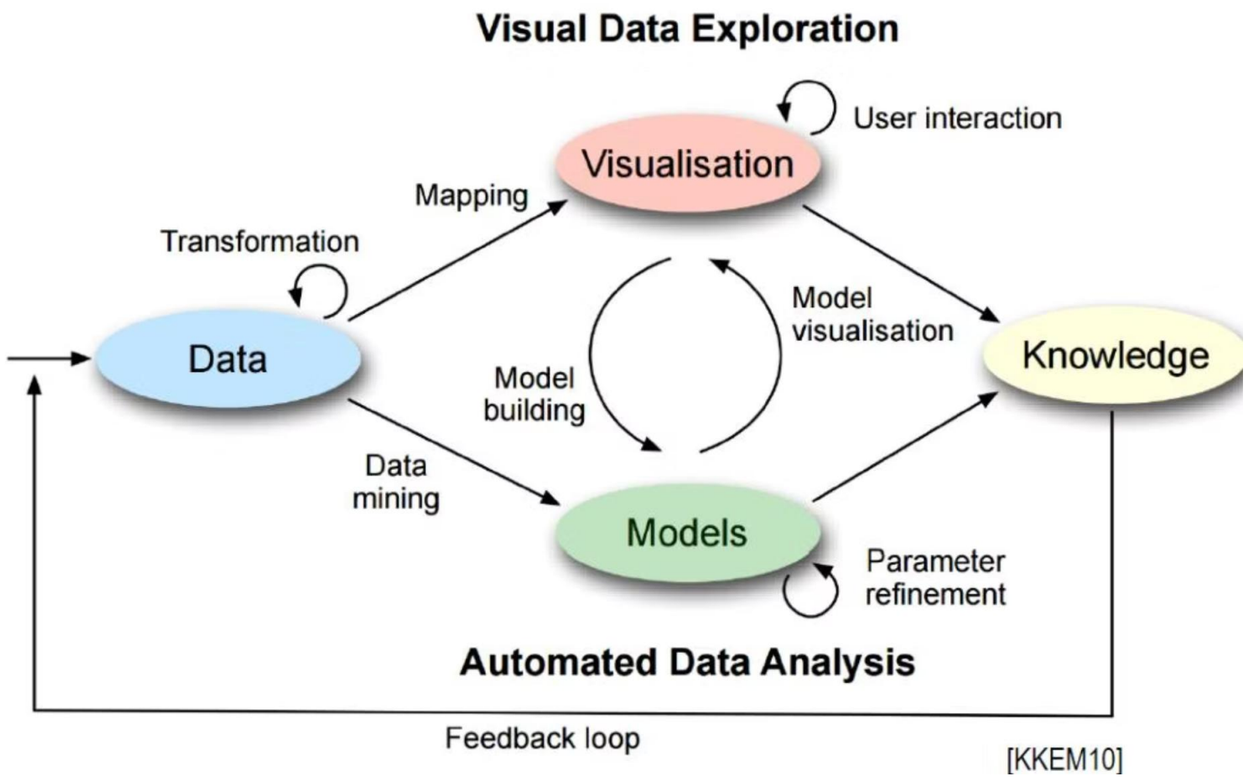
3

Developing or Improving Machine Learning Models

4

Conducting a Visual Analysis of Data and make deductions from it directly.

Visual Analytics Process



Visual Analytics Workflow by Keim (2010) [KKEM10]

Visual Data Exploration

Data → Transformation → Visualisation → Knowledge, with user interaction and model visualisation feedback loops.

Automated Data Analysis

Data → Data mining → Models → Knowledge, with parameter refinement and model building loops.

Feedback loop

A large feedback loop connects Knowledge back to Data, enabling continuous refinement.

What do we want you to do with Data Visualization? Which insights do we want you to generate?

How to use Data Visualization

1 "Rear Mirror"

Data Visualization is an easy and effective way to see how well you have cleaned your dataset.

2 Detecting Outliers

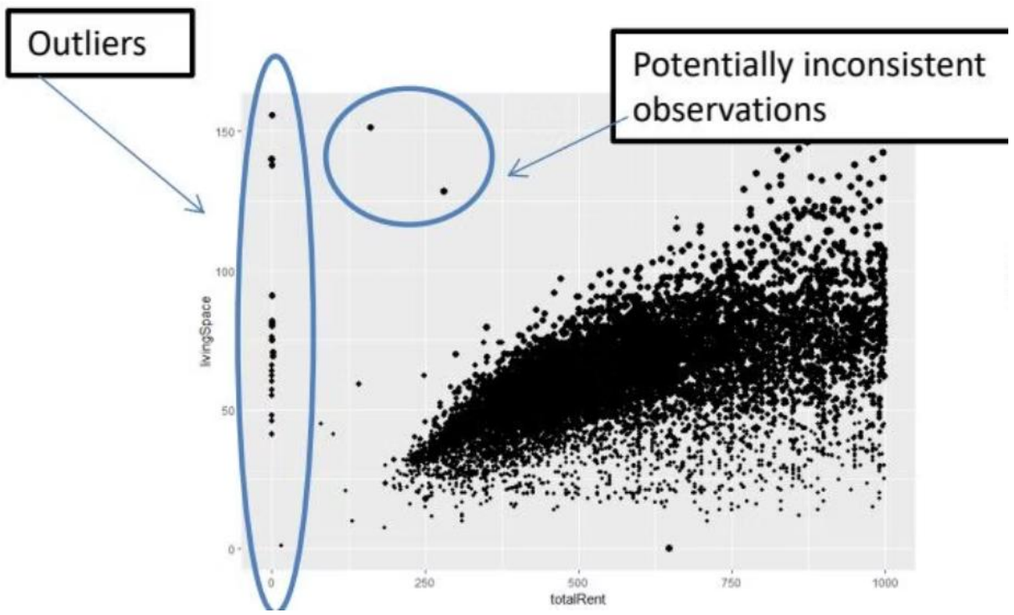
Histograms and scatter plots, for instance, can help you with detecting any remaining outliers.

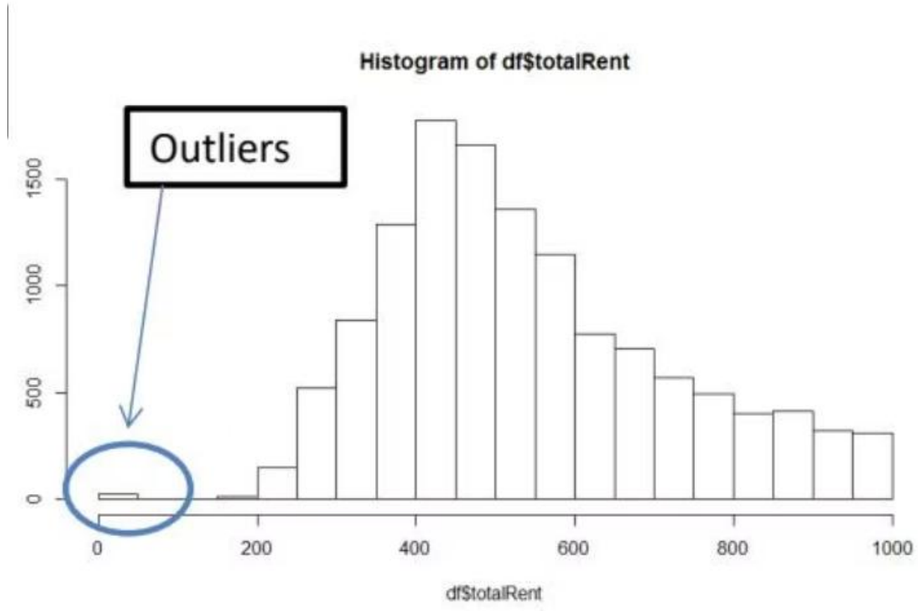
3 Spotting Inconsistencies

Scatter plots can give you indications about potential inconsistencies.

4 Looking ahead

Data Visualization can make relationships between variables visible.





How to use Data Visualization

→ **"Rear Mirror"**

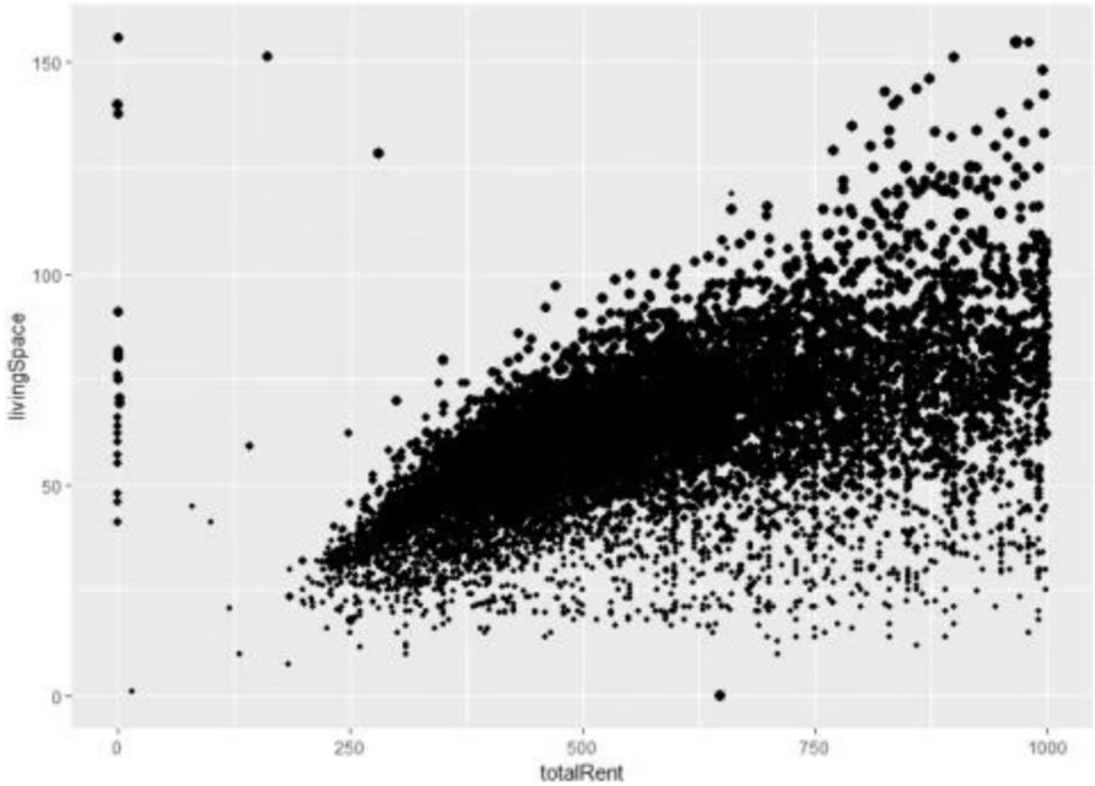
Data Visualization is an easy and effective way to see how well you have cleaned your dataset.

→ **Looking ahead**

Data Visualization can make relationships between variables visible.

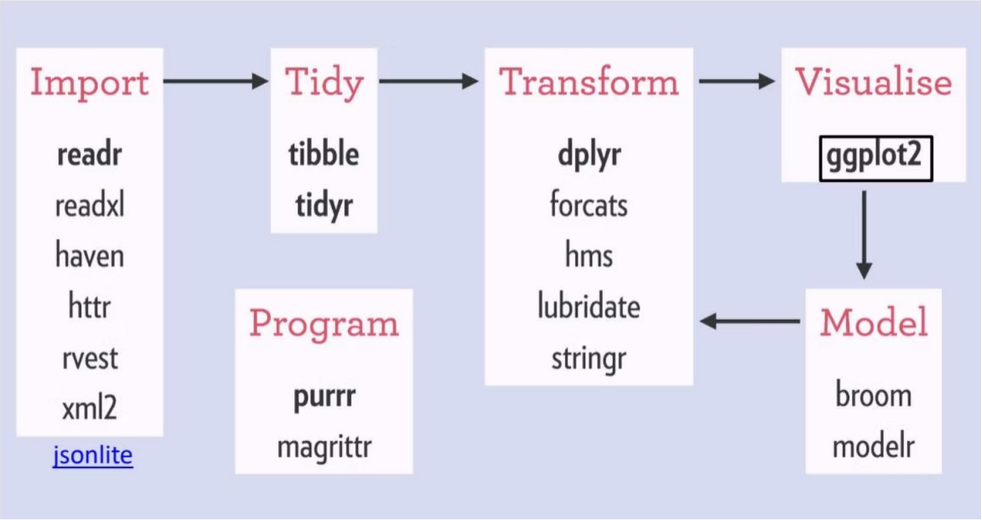
→ **Positive Relationship Revealed**

The scatter plot shows an upward relationship between living space and total rent.



Visualizing Data in R

Tidyverse



Import

readr, readxl, haven, httr, rvest, xml2, jsonlite

Tidy

tibble, tidyr

Transform

dplyr, forcats, hms, lubridate, stringr

Visualise

ggplot2

Program

purrr, magrittr

Model

broom, modelr

Visualizing Data in R

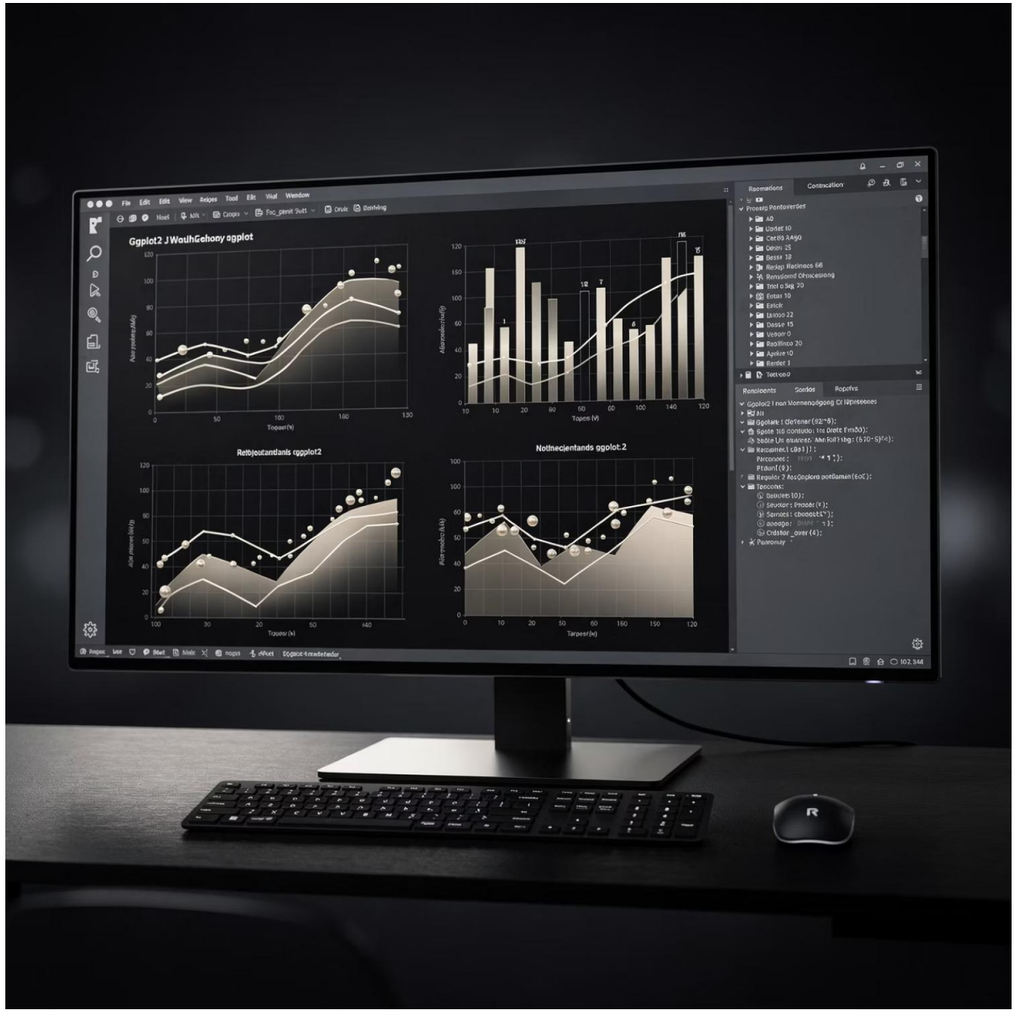
ggplot2 package

What is ggplot2?

Ggplot2 is an R package that helps you with exploring datasets visually. It is part of the tidyverse package that we have already seen earlier.

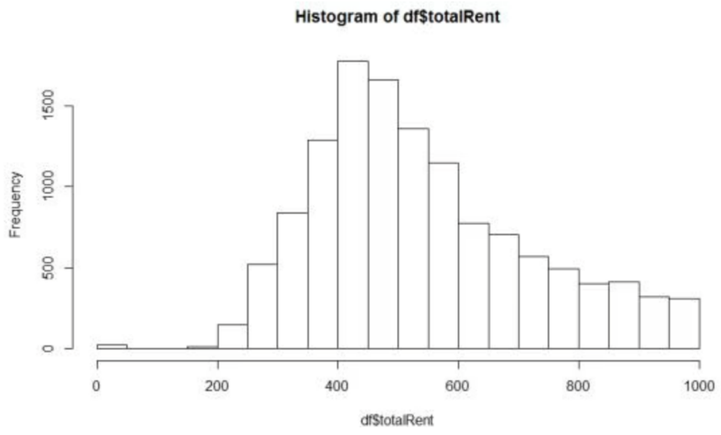
Cheat Sheet

https://github.com/rstudio/cheatsheets/blob/master/data-visualization-2.1.pdf



Getting an overview – Distribution Plots

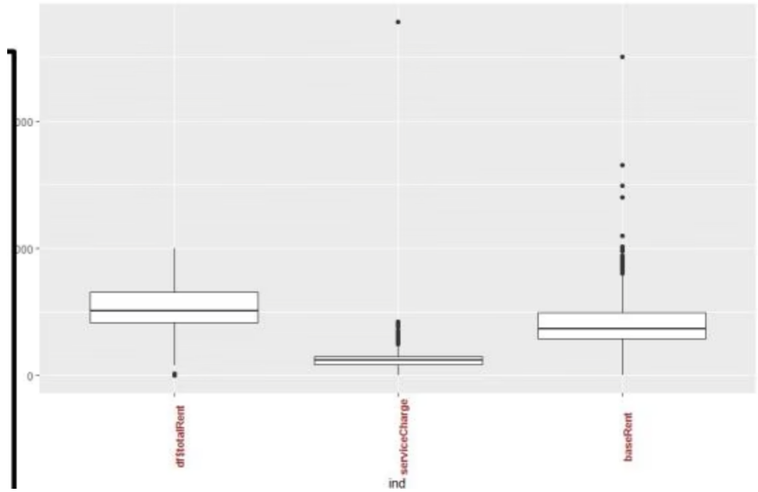
Histograms



A histogram titled **Histogram of df$totalRent**. The distribution is right-skewed, with a peak frequency of approximately 1500 at a rent value of 400. The frequency decreases as rent increases, with a long tail extending towards 1000.

```
# Histogram
> hist(df$totalRent)
```

Boxplots

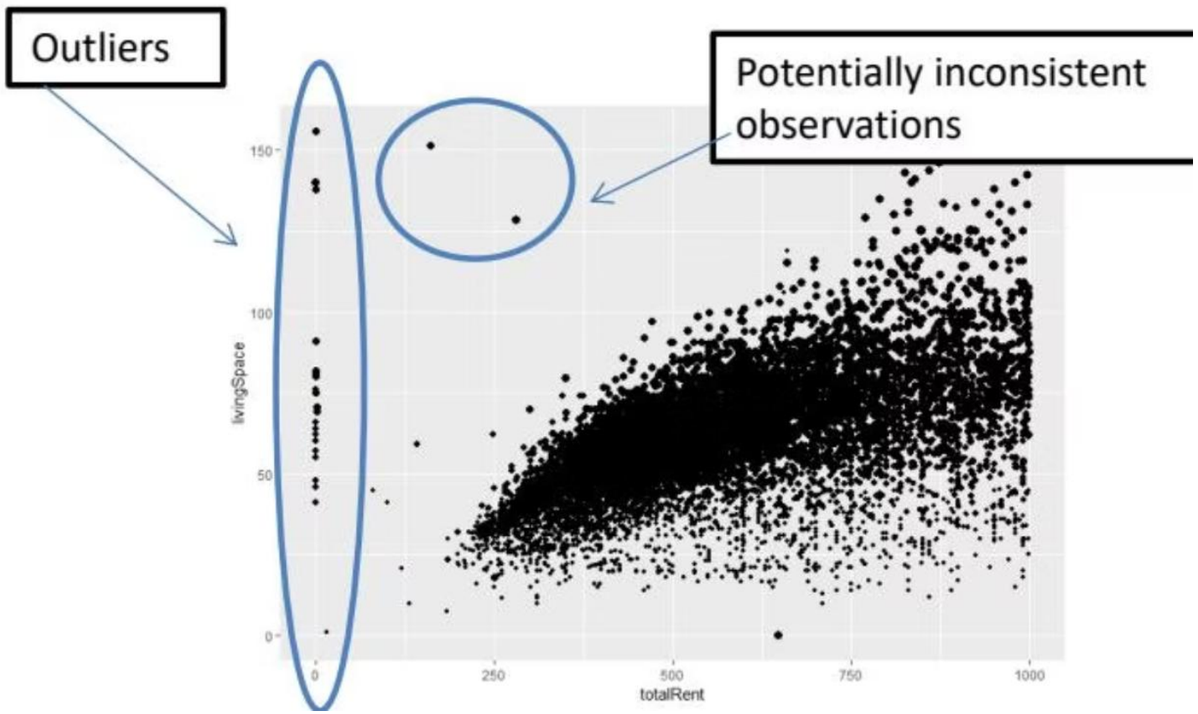


A boxplot comparing three variables: **df$totalRent**, **df$serviceCharge**, and **df$baseRent**. The **df$totalRent** box is the largest, spanning from approximately 100 to 900, with a median around 400. All three variables show some outliers.

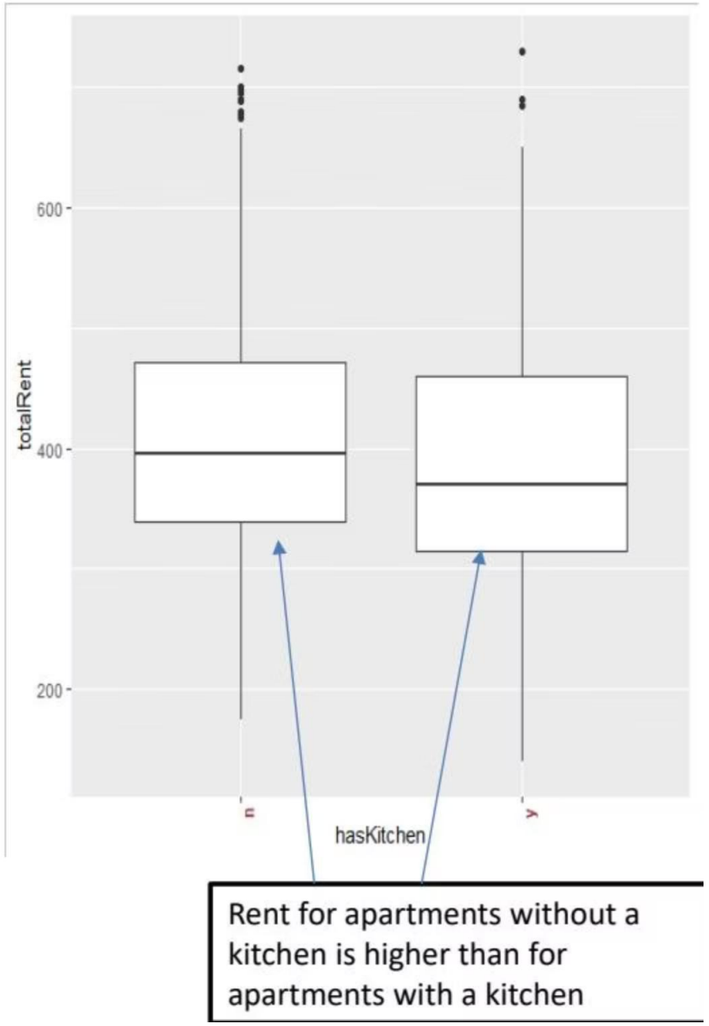
```
# Boxplot
> library(tidyverse)
> df.1 <- as.data.frame(df$totalRent)
> df.1$serviceCharge <- df$serviceCharge
> df.1$baseRent <- df$baseRent
> p <- ggplot(stack(df.1), aes(x = ind, y = values)) +
 geom_boxplot() +
 theme(axis.text.x = element_text(face = "bold",
 color = "#993333", angle = 90))
> p
```

Getting an overview

Sometimes, these visualizations reveal somewhat **counterintuitive relationships** or **mistakes**



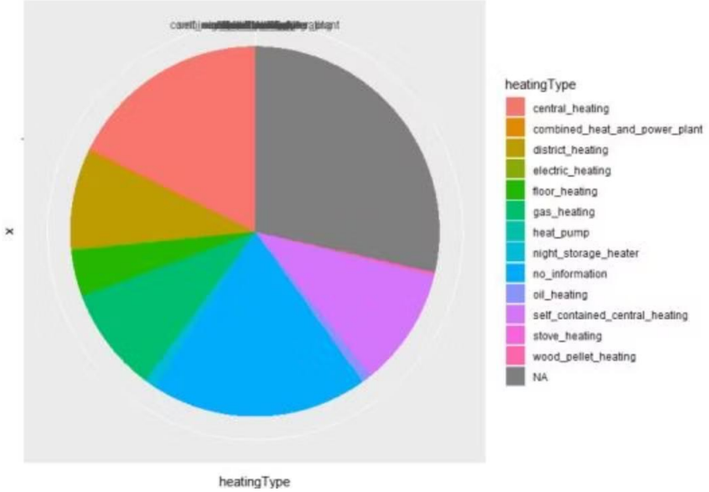
A scatter plot showing the relationship between **totalRent** (x-axis, 0 to 1000) and **livingSpace** (y-axis, 0 to 150). Annotations include: **Outliers** – a blue oval highlights a cluster of points with high living space and low rent. **Potentially inconsistent observations** – a blue circle highlights a few points with low living space and high rent.



A box plot showing **totalRent** (y-axis, 200 to 600) versus **hasKitchen** (x-axis, 0 and 1). **Rent for apartments without a kitchen is higher than for apartments with a kitchen**: A box at the bottom contains this text, pointing to the higher median rent for the group without a kitchen.

Getting an overview – Pie Charts

Histograms | **Boxplots** | **Pie Charts**



A pie chart titled **heatingType** showing the distribution of various heating types. **central_heating** is the largest segment (~25%), followed by **oil_heating** (~15%) and **district_heating** (~10%).

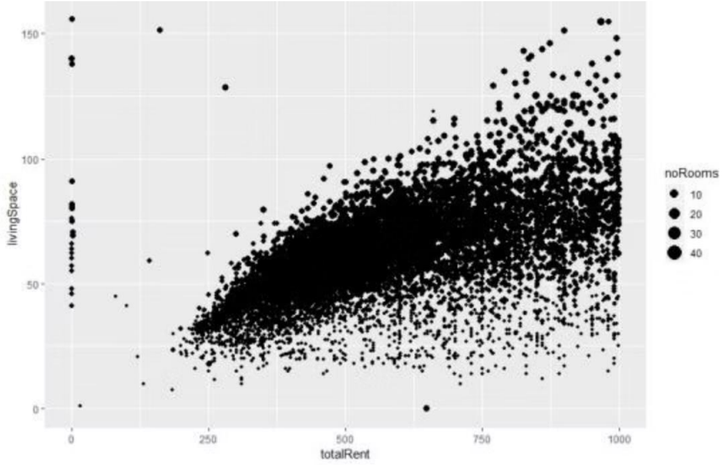
```
# Pie Charts
> library(tidyverse)
bp<- ggplot(df, aes(x="", y=heatingType,
  fill=heatingType))+
  geom_bar(width = 1, stat = "identity")
# bp
> pie <- bp + coord_polar("y", start=0)
> pie
```

heatingType	Approx. Proportion
central_heating	25%
district_heating	10%
gas_heating	5%
oil_heating	15%
stove_heating	5%
wood_pellet_heating	5%
combined_heat_and_power_plant	5%
electric_heating	2%
floor_heating	2%
heat_pump	2%
night_storage_heater	2%
no_information / NA / other	~3%

Zooming in – Relationships between variables

Scatter Plots

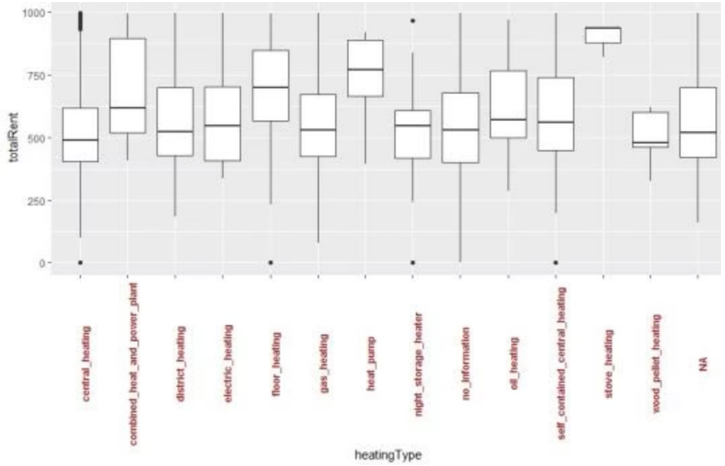
```
> library(tidyverse)
# Scatterplot
> scatter <- ggplot(df, aes(x=totalRent,
  y=livingSpace)) +
  geom_point(aes(size=noRooms))
> scatter
```



A scatter plot showing the relationship between **totalRent** (x-axis, 0 to 1000) and **livingSpace** (y-axis, 0 to 150). The plot includes a legend for **noRooms** with sizes corresponding to 10, 20, 30, and 40. The data points show a positive correlation, with larger points representing more rooms.

Box Plots

```
# Boxplot
> p <- ggplot(df, aes(x=heatingType,
  y=totalRent)) +
  geom_boxplot() +
  theme(axis.text.x = element_text(face = "bold",
  color = "#993333", angle = 90))
> p
```



A box plot showing the distribution of **totalRent** for different heating types. The heating types are listed on the x-axis. The y-axis represents totalRent from 0 to 1000.

Zooming in – Relationships between variables

Scatter Plots

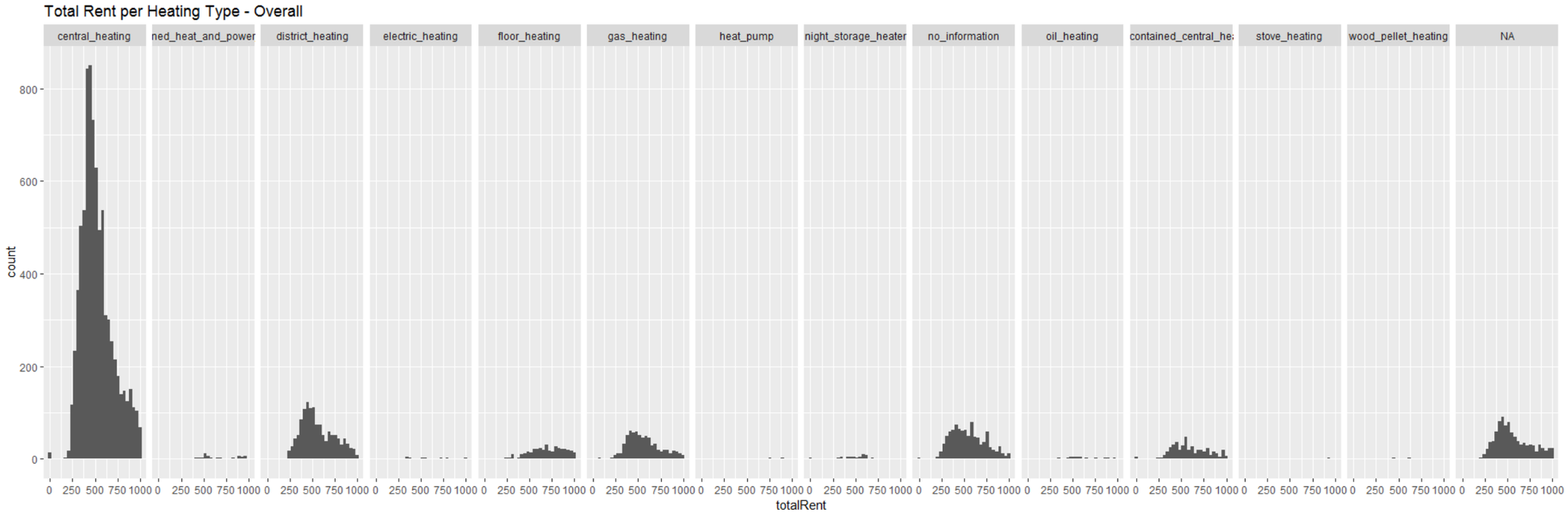
Box Plots

Bar Charts

```
> library(tidyverse)

# Scatterplot
> k <- ggplot(df, aes(totalRent, stat(count))) +
  geom_histogram(binwidth = 0.2) +
  geom_histogram(aes(y = stat(count))) +
  ggtitle("Total Rent per Heating Type - Overall")
> k + facet_grid(. ~ heatingType)
```

Total Rent per Heating Type – Overall. A series of 15 histograms showing the distribution of total rent for different heating types. The x-axis is **totalRent** ranging from 0 to 1000, and the y-axis is **count** ranging from 0 to 800. The **central_heating** plot is the most prominent, showing a high frequency of rents between 500 and 750. Other plots show much lower frequencies, generally below 200.



Questions?